

- 1.) Paper 1 Summary: J. S. Herd and M. D. Conway, "The Evolution to Modern Phased Array Architectures," in Proc. of the IEEE, vol. 104, no. 3, pp. 519-529, March 2016

This article outlines the evolution of phased array technology developed over the last 60 years. The article introduces the reader first to the transition from mechanically steered beams to electronic beam steering in the mid-1940's. The article outlines the three major categories of phased arrays as passive arrays, active arrays, and digital arrays. Passive arrays are defined as having an array fed by a centralized high power transmit amplifier with a low-loss power combining network accompanied by a phase shifter at every array element. Active arrays are defined as having the amplification stages, both transmit and receive modes, distributed at each array element. Digital arrays are defined as having the RF frequencies down converted to digital signals at the array element level with beamforming performed digitally with a digital signal processing (DSP) technique.

In 1958 the Advanced Research Projects Agency (ARPA) began to develop Electronically Steered Array Radar (ESAR) technology which utilized high power transmit amplifier tubes along with low-cost mixers and phase shifters. This effort was for the development of defense system radars for military applications. In 1980 Raytheon built the PAVE PAWS radar which was the first solid-state phased array radar. Solid-state refers to the high power transmit amplifiers in the array as being integrated IC components as opposed to traveling wave tube (TWT) amplifiers. Large missile defense and space surveillance radars were developed for the Department of Defense (DoD) by Raytheon which included the SPY-1A (S-band) for the Navy and the Patriot (C-band) for the Army. Lockheed Martin developed the Joint Surveillance Target Attack Radar System (J-STARS) (X-band) for the U.S. Air Force.

Beginning in 1988 through 1995 the Defense Advanced Research Projects Agency (DARPA) developed the Microwave and Millimeter Wave Monolithic Integrated Circuits (MIMIC) program. This program developed manufacturing processes for gallium arsenide (GaAs) devices using the model-based systems engineering (MBSE) approach. This allowed for multiple design iterations to be modeled before going into production. Along with this, DARPA developed unique packaging techniques in order to allow GaAs MMIC's to be a feasible solution for high power amplifiers and IC's. From 1995 to 1999 DARPA and the Microwave and Analog Front End Technology (MAFET) program deployed the new simulation-based methodology which aimed to reduce production time. This inevitably led to deployment of active phased array technology in the 1990's.

Going into the new millennia, DARPA Wide Band Gap Semiconductor for RF Applications (WBGs-RF) initiated a new area of research in Gallium Nitride on Silicon Carbide (GaN-on-SiC) technology. This allowed for higher power amplifiers with greater power-added efficiency than previous GaAs technology. Along with these higher powers came a cost of efficiency over time due to a rise in MMIC temperatures. In 2012 DARPA Intra/Interchip Enhanced Cooling (ICECool) program developed embedded cooling solutions for high power devices. These advancements led to the proficiency of active phased arrays for long-term military operations. Along with these improvements, Raytheon led the development of T/R modules which are sub-systems of a transmit and receive chain controlled by a common-leg or switching circuit. These modules utilized digital

beamforming with the use of analog beamforming, down-converters, analog-to-digital converters (ADC), and digital processors to manipulate the incoming signals. This allowed for multiple simultaneous beams over a limited scan angle. While the cost of digital receiver-exciter (DREX) components goes down, element-level digital beamforming is possible. This allows for preservation of aperture information as well as a digital manipulation of signals through discrete states rather than received signal power. Another upside to this is the arbitrary phase shift time delay of the digital phase shifters which allows large arrays to scan over wide bandwidths without degrading signal integrity.

With element-level beamforming networks, cost per component becomes a crucial point of interest. Typical phased array radar systems that utilize the element-level beamforming are configured in a slat array system. These slats are arranged perpendicular to the array aperture face. This configuration allows for easier thermal management and maintenance as the slats are modular. With this comes the price of RF boards and cables for each of the slats. With large aperture areas the number of channels will be in the thousands which leads to a high cost. Alternatively there is the tile configuration which has antenna elements with T/R modules connected to a shared multilayer RF board. This drives the cost of materials down for the array, however, thermal management is harder to realize with this configuration.

The article concludes with an outlook of future technology involving GaN 3D bonded to Si CMOS wafers. Doing this allows for high power, high speed, and low cost per chip. As the young technology of solid-state transmitter design matures, the development of active phased arrays does so as well.

- 2.) B. Sadhu, X. Gu and A. Valdes-Garcia, "The More (Antennas), the Merrier: A Survey of Silicon-Based mm-Wave Phased Arrays Using Multi-IC Scaling," in IEEE Microw. Mag., vol.20, no. 12, pp. 32-50, Dec. 2019

This article starts by talking about how phased array antennas began. Karl Ferdinand was the first to develop these antennas in 1905 in 1977 during the Cold War the invention of phased array radars for ballistic missile early warning systems such as the Cobra Dane was developed. In 1980 the precision acquisition vehicle entry phased array warning system was developed as well as the Sea-Based X-Band Radar-1 (SBX-1). Si-germanium (SiGe) bipolar CMOS and Si CMOS technology was first developed at the California Institute of Technology (Caltech). Until then phased array radar systems were too expensive for commercial applications. Significant research was done on Si based millimeter-wave phased array technology. This drove down the cost of integrated chips that operate at high frequencies dramatically. Along with this was the limitation to the number of signal chains within an integrated circuit. Even for today's current technology this limit is 64 signal chains. This limit has been reached first by IBM with Intel and the University of California San Diego to follow. This method of creating large, phased arrays is called phased-array scaling. The benefits of phased-array scaling include driving yield up, lowering the cost of identical integrated circuits, testing multiple units can be automated easily, scalability to tile size allows for flexibility in design.

With the current technology, beamforming can be performed at multiple points in a phased-array chain. RF beamforming requires the least amount of components which includes RF phase shifting, frequency conversion, analog-to-digital conversion, and digital signal processing functions. All of these functions can be broken up into individual unique circuits with the downside of a difficult RF connection between the RF beamforming integrated circuits and RF-to-IF down-conversion circuits. The article then goes on to talk about the importance of local oscillator (LO) phase variation that may result in phase error of the chain. These local oscillators need to be synchronized in order to support various chains. This is mitigated by the use of phase-locked loops (PLL) which uses a reference signal to sync the input phase. The PLL circuit also synchronizes the mm-wave frequency at the output. These, of course, have their drawbacks as PLL circuits require a voltage-controlled oscillator which can add phase noise that is difficult to track versus correlated phase noise.

The next section of the article talks about the implementation of antenna arrays and the packaging of components. For phased-array systems, an array element spacing of $\lambda/2$ is required with λ being the wavelength of interest. Various packaging techniques include antennas on PCB, antennas on package with embedded antennas, and antennas on a chip or wafer glass substrate with a second-level PCB. As frequency goes up, the wavelength will go down which constricts the antenna element spacing. This makes fitting the integrated chips into the package spacing a problem of element arrangement. Most inductor-based designs use an N-element matrix package scheme which has the integrated circuits arranged in an evenly spaced arrangement. As frequency goes higher in the millimeter-wave frequencies, transmission line antenna designs are implemented in a $M \times N$ matrix. As a solution to the packaging challenges at high frequencies, IBM developed dummy antennas with

no electronics connected, allowed the proper antenna spacing to work above 60 GHz. The ratio of active antenna area to total array area is called the fill factor. IBM used a fill factor of 64% with the drawback of grating lobes that will distort the side lobes of the directive antenna gain.

The article concludes by talking about multiple future considerations with Si based antenna phased-array systems. Beamforming, scan angle, and bandwidth are always a consideration when determining the speed of operation of the radar. With high power electronics, tightly packed together operating at high frequencies, thermal management is crucial. This will have a drastic impact on performance of the array which will limit the overall size. As technology continues to push the limits of integrated chip efficiencies, power lost due to heat will be minimized and larger arrays can be realized. Software plays a critical role in phased-array design, modeling, and function. As software algorithms develop, so will the ability to control phased-array systems. Integrated chips are now becoming digitally controlled while using the same high frequency materials. There are clear challenges when developing high millimeter-wave frequency phased-array systems, however, as the methods and concepts explained in this article continue to develop, so will our ability to communicate via the electromagnetic spectrum.

3. Array Element Channel Gain, Noise Figure, and IIP3 Calculator

Given Cascade:

- LNA ($gain = 12dB \pm 1dB$, $VSWR_{in} = 1$, $VSWR_{out} = 1.5$, $NF_{mean} = 2dB$, $OIP3 = 0dBm$)
- Interconnect 1 ($loss = 1.5dB$, $VSWR_{in} = 1.5$, $VSWR_{out} = 1.5$)
- VGA ($gain = 8dB \pm 2dB$, $VSWR_{in} = 1$, $VSWR_{out} = 2$, $NF_{mean} = 4dB$, $OIP3 = 10dBm$)
- Attenuator ($loss = 8dB \pm 0.5dB$, $VSWR_{in} = 2$, $VSWR_{out} = 1.5$)
- Gain Block ($gain = 2dB \pm 2dB$, $VSWR_{in} = 1.5$, $VSWR_{out} = 2.8$, $NF_{mean} = 3dB$, $OIP3 = 10dBm$)
- Interconnect 2 ($loss = 0.8dB$, $VSWR_{in} = 2.8$, $VSWR_{out} = 3.2$)
- Gain Block ($gain = 15dB \pm 2dB$, $VSWR_{in} = 3.2$, $VSWR_{out} = 1$, $NF_{mean} = 5dB$, $OIP3 = 24dBm$)

Table 1: Excel Cascade Gain, TOI, and Noise Figure Calculator

Circuit 1											
Device No.	1	2	3	4	5	6	7				
Device Type	LNA	Interconnect 1	VGA	Attenuator	Gain Block 1	Interconnect 2	Gain Block 2				
G (dB)	12.00	-1.5	8	-8	2	-0.8	15				
OTOI (dBm)	0.00		10		10		24				
G(power)	15.85	0.694	6.31	0.141	1.96	0.76	22.95	<-- Values are with VSWR Mismatch			
OTOI (W)	0.001	0.001	0.010	0.001	0.010	0.001	0.251				
ΣOTOI (W)	0.001	0.0007	0.003	0.0004	0.0008	0.0006	0.0129				
ΣOTOI (dBm)	0.00	-1.59	4.83	-3.67	-1.09	-2.26	11.12				
ΣG (dB)	12.00	10.41	18.41	9.91	12.84	11.67	25.27				
ΣITOI (dBm)	-12	-12.00	-13.58	-13.58	-13.93	-13.93	-14.16				
	User Defined		VSWR In	Gamma In	VSWR Out	Gamma Out					
F1 (dB)	2	LNA1	1	0	1.5	0.2		<-- Change values in this table to see code run			
G1(dB)	12	LNA1									
L1 (dB)	1.5	Interconnect 1	1.5	0.2	1.5	0.2					
F2 (dB)	4	VGA	1	0	2	0.33					
G2(dB)	8	VGA									
L3 (dB)	8	Attn	2	0.33	1.5	0.2					
F3 (dB)	3	Gain Block 1	1.5	0.2	2.8	0.47					
G3 (dB)	2	Gain Block 1									
L4 (dB)	0.8	Interconnect 2	2.8	0.47	3.2	0.52					
F4 (dB)	5	Gain Block 2	3.2	0.52	1	0					
G4 (dB)	15	Gain Block 2									
	Component	G (dB)	G	L (dB)	L	F (dB)	F	sum L	sum G to Tsky	F	Sum f
										290	290
	0dB Loss	0	1	0	1	0	1	1	1	0	0
matched	LNA1	12	15.85	0	1	2	1.585	1.000	15.85	1.585	1.585
	Interconnect 1	0	0.694	1.5	1.413	1.5	1.442	1.413	10.992	0.028	1.613
matched	VGA	8.0000	6.3096	0.0000	1.0000	4.0000	2.5119	1.413	69.3532	0.138	1.750
	Attn	0.0000	0.141	8.0000	6.3096	8.0000	7.078	8.913	9.7978	0.088	1.838
	Gain Block 1	2.0000	1.9616	0.0000	1.0000	3.0000	2.2441	8.913	19.2198	0.127	1.965
	Interconnect 2	0.0000	0.764	0.8000	1.2023	0.8000	1.309	10.72	14.6778	0.016	1.981
	Gain Block 2	15.0000	22.9462	0.0000	1.0000	5.0000	5.5408	10.72	336.8000	0.309	2.290
									Total NF (dB)		3.5991
									Cumulative Gain (dB)		25.2737

Table 2: Cumulative Values Matched vs. Mismatched Components

At a Glance				
	Min	Typical	Max	Matched
Cumulative NF (dB)	4.90	3.60	2.91	3.15
Cumulative Gain (dB)	18.78	25.27	31.77	26.70
Σ OTOI (dBm)	6.46	11.12	15.22	13.92
Σ G (dB)	18.78	25.27	31.77	28.30
Σ ITOI (dBm)	-12.32	-14.16	-16.55	-14.38

Table 1 shows the calculated values for OTOI, cumulative Gain and Noise Figure for the given cascade. The EXCEL code performs operations based on values the user inputs into the 'User Defined' table. A check is performed for cumulative gain with the two tables it produces for cascaded OTOI and Noise Figure. We can compare the cumulative gain values highlighted in green (25.27) and magenta (25.27). In order to manipulate the code the user can change the 'User Defined' table and then formulas embedded into the cascaded TOI and NF tables with call on those values.

Table 2 shows the matched vs. mismatched cascade using given VSWR values for input and output ports. When looking at the typical values there is an increase in noise figure with a decrease in gain. This is to be expected as mismatched components will lead to an increase in insertion loss with an increase in noise power and therefore noise figure. A decrease in OTOI can also be observed with the mismatched case. This translates to less power output at compression.

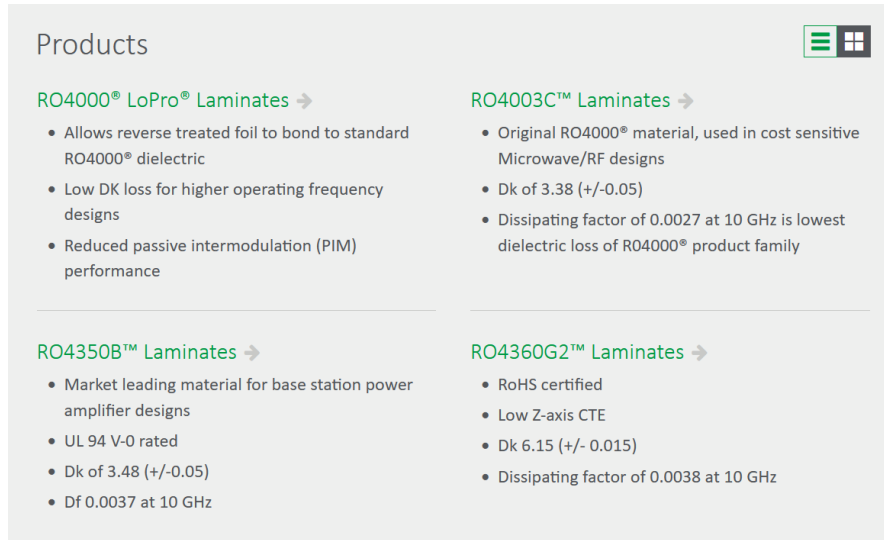
4. Design Challenge: Design a receive chain with the following parameters:

- Frequency Range: 27 – 30 GHz.
- Commercial off-the-shelf (COTS) components
- Components are limited to: LNA, Gain Blocks, Variable Attenuator, Variable Phase Shifter, and Filter
- Interconnects should be microstrip designed with electrical length no shorter than 0.2λ at highest frequency of operation (30 GHz.)
- Standard laminant board thickness:
 - 31 mil (~0.8 mm) board thickness
 - 1oz. (35 μ m) Copper (Cu) cladding – Conductivity of Cu: 5.96×10^7

I. Microstrip Design

Keeping to standard PCB dimensions, three different dielectric laminates were selected for a trade study of varying properties. These laminates are standard FR4, Rodgers 4003C, and Rodgers 4350B LoPro. FR4 has been the standard for high frequency PCB manufacturing due to its low cost and availability. The Rodgers boards are specialized to custom designs with each providing unique benefits and costs. Costs for these specialty boards are non-standard which makes accounting for their cost difficult. Figure 1 shows the different Rodgers laminates.

Figure 1: Rodgers Dielectric Laminates



Products	
RO4000® LoPro® Laminates →	RO4003C™ Laminates →
• Allows reverse treated foil to bond to standard RO4000® dielectric • Low DK loss for higher operating frequency designs • Reduced passive intermodulation (PIM) performance	• Original RO4000® material, used in cost sensitive Microwave/RF designs • Dk of 3.38 (+/-0.05) • Dissipating factor of 0.0027 at 10 GHz is lowest dielectric loss of RO4000® product family
RO4350B™ Laminates →	RO4360G2™ Laminates →
• Market leading material for base station power amplifier designs • UL 94 V-0 rated • Dk of 3.48 (+/-0.05) • Df 0.0037 at 10 GHz	• RoHS certified • Low Z-axis CTE • Dk 6.15 (+/- 0.015) • Dissipating factor of 0.0038 at 10 GHz

RO4003C was chosen as one of the more cost sensitive options while the RO4350B LoPro was chosen for its low dielectric constant at higher frequencies. These substrates were compared to FR4 in microstrip design.

Table 1: Dielectric Trade Study

Microstrip Dielectric Trade Study					
	ϵ_r	TanD	Dielectric Thickness (mm)	Conductor Thickness (μm)	Conductivity
FR4	4.4	0.017	0.780	35	5.96E+07
RO4003C	3.55	0.0027	0.813	35	5.96E+07
RO4350B LoPro	3.55	0.0037	0.780	35	5.96E+07

Microstrip lines were designed and simulated using Keysight ADS and the built-in LineCalc tool. This tool uses given dielectric properties at a given frequency, impedance, and electrical length. For this design, the desired frequency range is 27 – 30 GHz. with a center frequency of 28.5 GHz. All microstrip lines were designed at 30 GHz. with a quarter wavelength electrical length, matched to 50 Ohms. Figure 2 shows an example of the LineCalc tool as well as an example of a microstrip line simulation in ADS. Figure 4 shows the comparison between FR4, RO4003C, and RO4350B LoPro small-signal response. To ensure that the microstrip electrical length meets the required specification, Figure 5 shows the phase of the microstrip at 30 GHz.

Figure 3: Keysight ADS Microstrip Simulation and LincCalc Tool

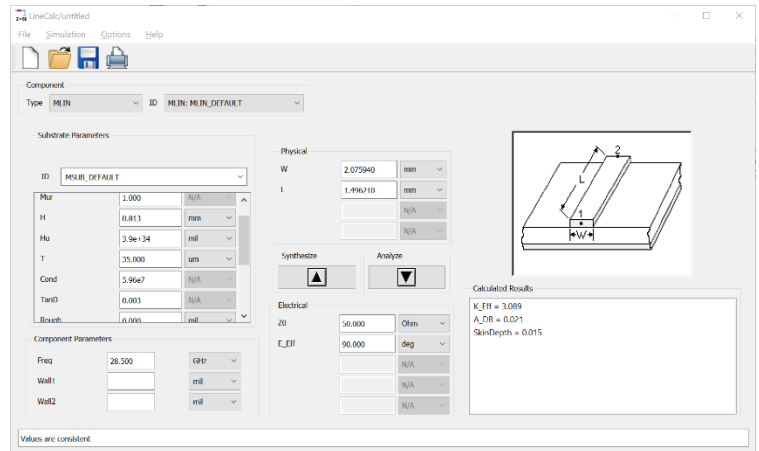
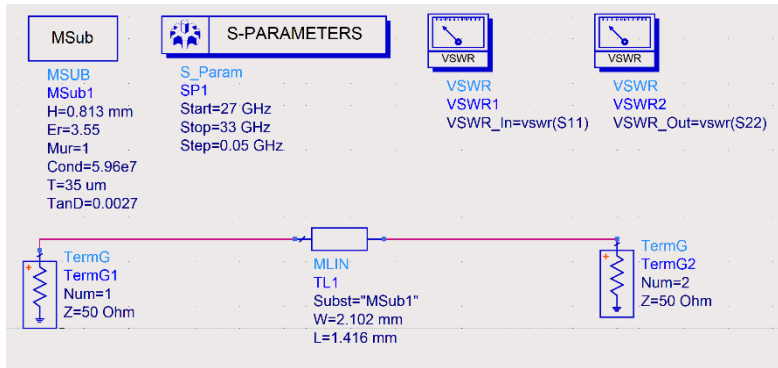


Figure 4: Microstrip Small-Signal Comparison

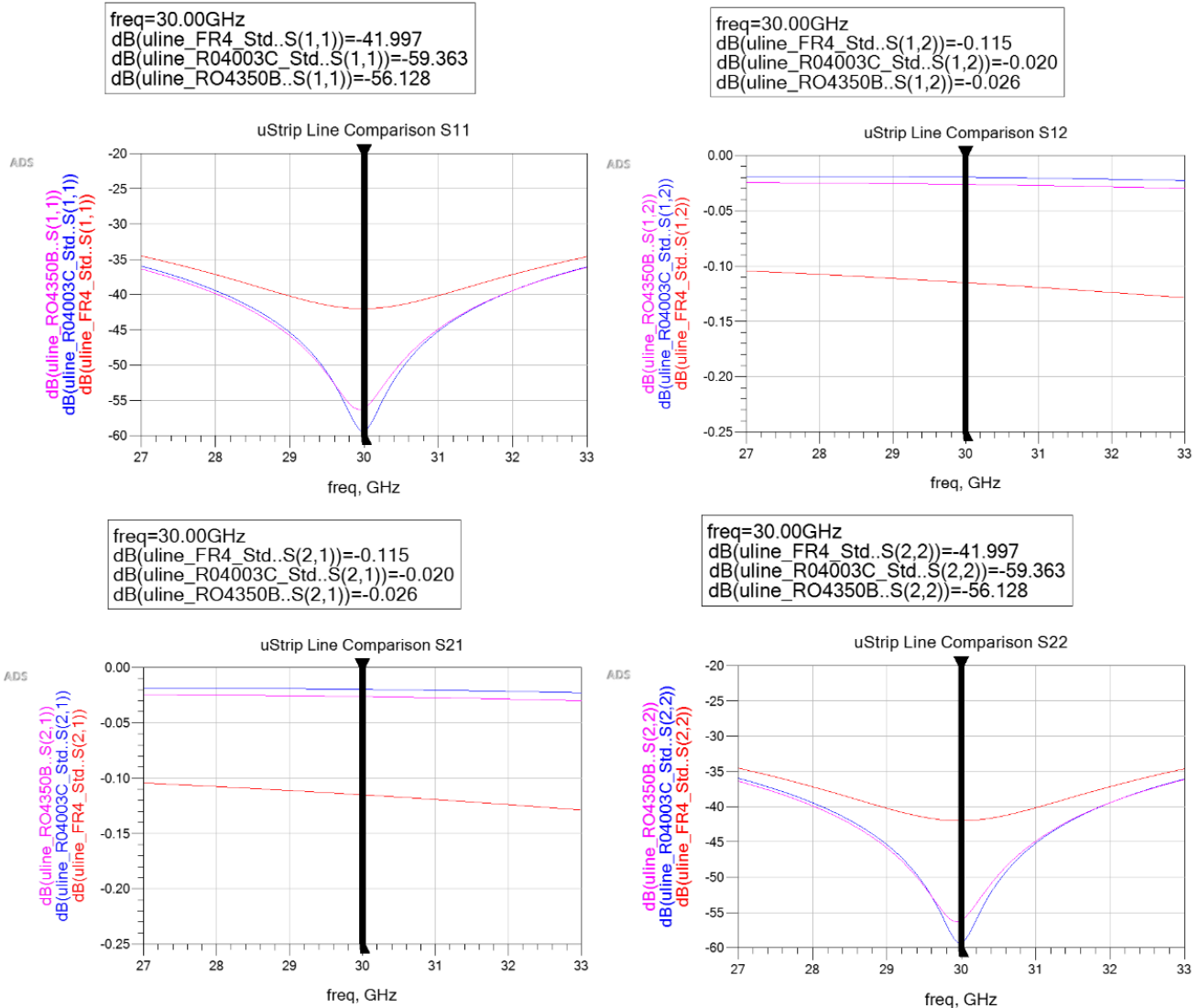
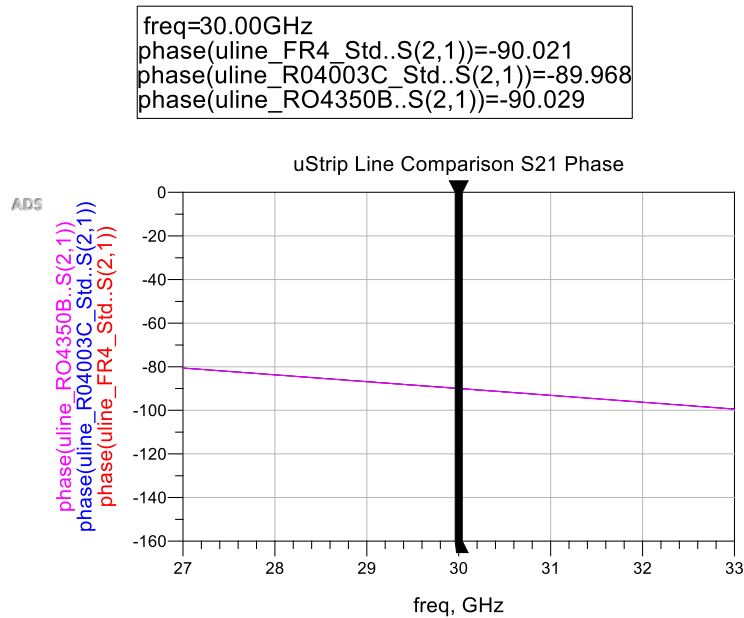


Figure 5: Microstrip Phase



Based on these results, the Rogers RO4300C board laminate was selected for the microstrip design used in the cascade model.

II. Commercial Off-the-Shelf (COTS) Component Trade Study

Initial inspection of the commercial market from commercial vendors such as Analog Devices, MACOM, and Qorvo prove that finding components in the 27-30 GHz. range was a challenge. Devices were selected based on requirements, availability, and cost. In order to keep to the frequency requirements of the design, a microstrip filter was designed using a utility called RF-Tools by Marki Microwave. This was done due to the lack of RF band pass filters for 27-30 GHz.

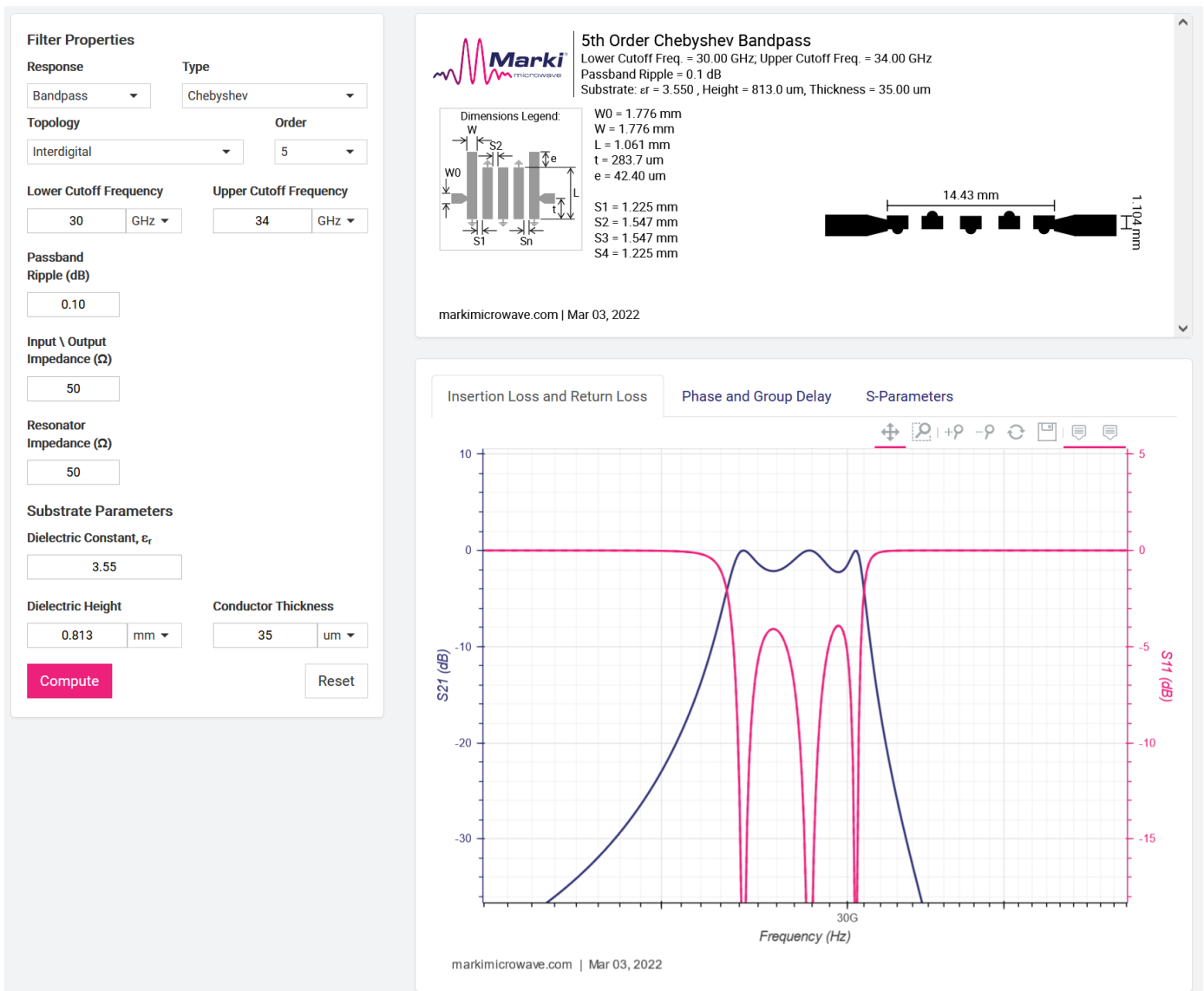
Components selected along with cost are shown in table 2. The cost was either determined from vendor websites or estimated for custom design. Figure 2 shows the calculated cascade parameters from the code written for problem 3. These values will be compared to the simulated values.

Table 2: Bill of Materials

Bill of Materials	Manf	Part No.	Est. Cost (Estimated)	Gain (dB)	OIP3 (dBm)	NF (dB)	Frequency (GHz.)	Package
LNA	Qorvo	CMD299K4	\$ 79.53	16d	18	4	18 - 40	SMT
5-bit Digital Attenuator	Qorvo	TGL2223-SM	\$ 76.82	-8.5	26	--	0.1 - 31	SMT
Gain Block 1	MACOM	MAAM-011112	\$ 21.22	24	30	6.5	20 - 37	SMT
Phase Shifter	Qorvo	TGP2100	\$ 195.00	-7	--	--	28 - 32	SMT
Standard FR4 Dielectric Board	Various	--	\$ 30.00	--	--	--	30	--
Rodgers RO4003C	--	--	\$ 60.00	--	--	--	30	--
Microstrip Filter (RF-Tools by Marki Microwave)	--	--		-0.1	--	--	30 - 34	--
Interconnects (Microstrip Line)	--	--		-0.1	--	--	30	--
Total		FR4 Board	\$ 402.57					
		RO4003C Board	\$ 432.57					

Components were selected for surface mount (SMT) connection with microstrip interconnects. Interconnects are assumed to be bonded or soldered to components. See BOM page for links.

Figure 6: Marki Microwave RF-Tool Used for Microstrip Bandpass Filter

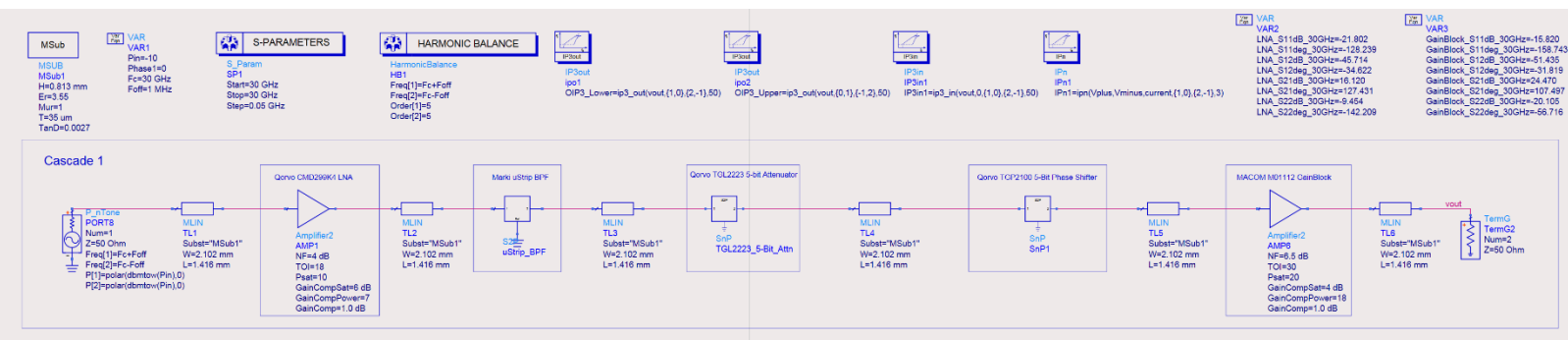


Filter dielectric constant, impedance, and conductor thickness are consistent with the microstrip designed at 30GHz. Although the lower and upper cutoff frequencies are different than the specified band, the results work in favor of the design specification of 27-30GHz. The filter's .s2p file was exported from the tool into ADS.

Devices were modeled in Keysight ADS using given s-parameter files as well as modeled non-linearly with data from available product data sheets. The LNA and Gain Block amplifier models contained non-linear data such as gain compression, noise figure, and third-order intercept. Components were arranged in three unique receive chain cascades. Each cascade was modeled at 27 GHz. and 30 GHz. for gain, noise figure, and third-order intercept. To ensure the frequency specification was met, small-signal response was simulated over a range of frequencies and cumulative gain was measured.

III. ADS Simulation Results

Figure 7: Cascade 1 Schematic



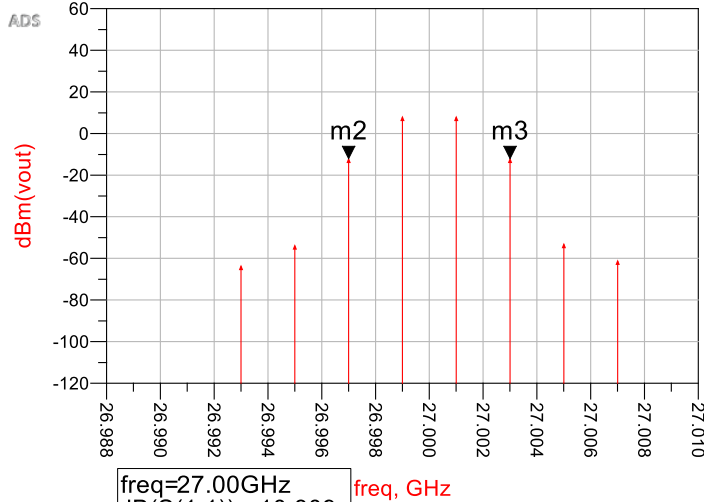
See schematic page for larger view.

Figure 8: Cascade 1 Third-Order Intercept and Cumulative Gain

m2
freq=27.00GHz
dBm(vout)=-12.071

m3
freq=27.00GHz
dBm(vout)=-12.092

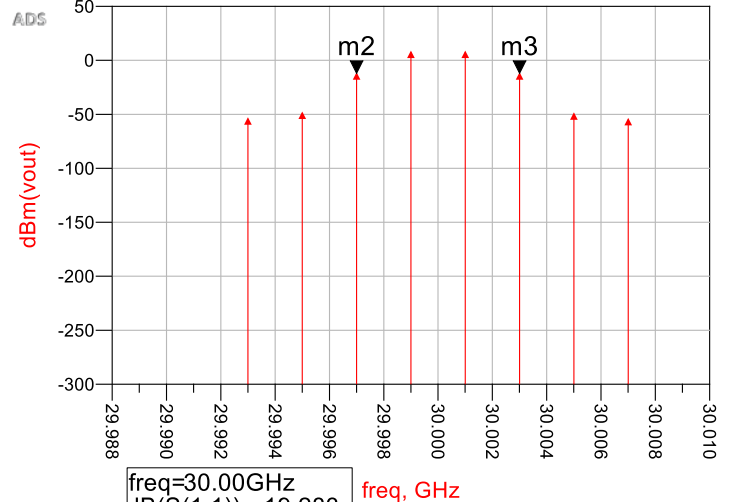
Cascade 1 Fundamental and Intermodulation Products 27GHz



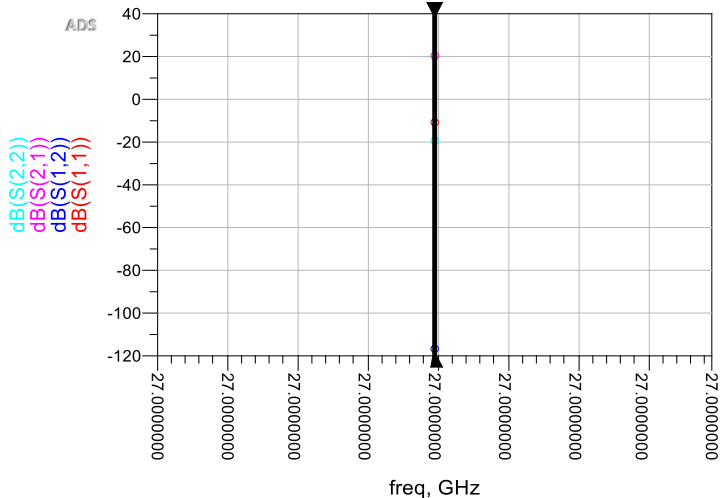
m2
freq=30.00GHz
dBm(vout)=-12.016

m3
freq=30.00GHz
dBm(vout)=-12.003

Cascade 1 Fundamental and Intermodulation Products 30GHz



Cascade 1 SS 27GHz



Cascade 1 SS 30GHz

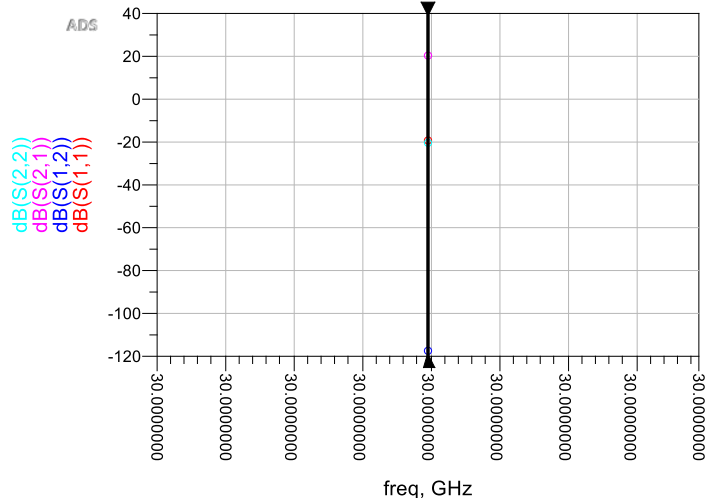


Figure 9: Cascade 2 Schematic

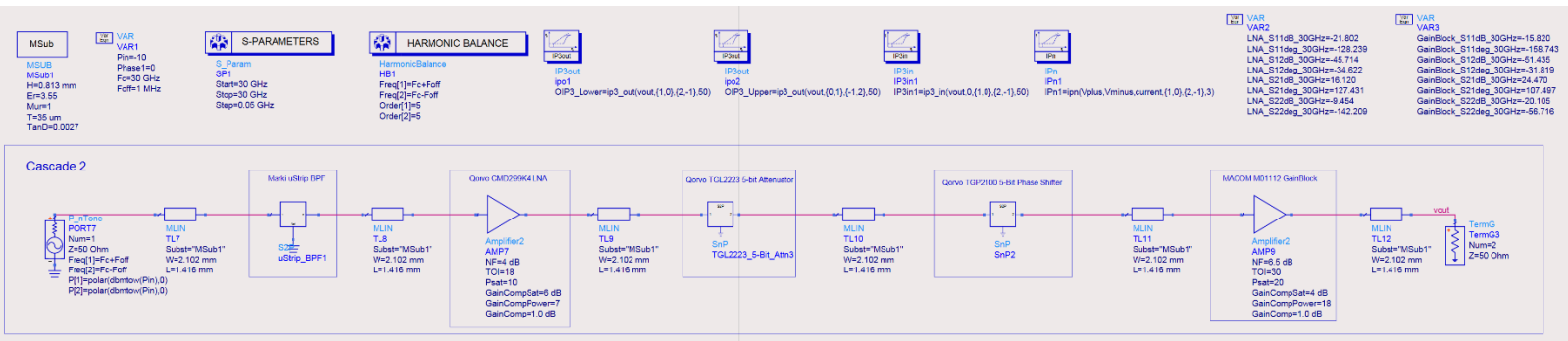


Figure 10: Cascade 2 Third-Order Intercept and Cumulative Gain

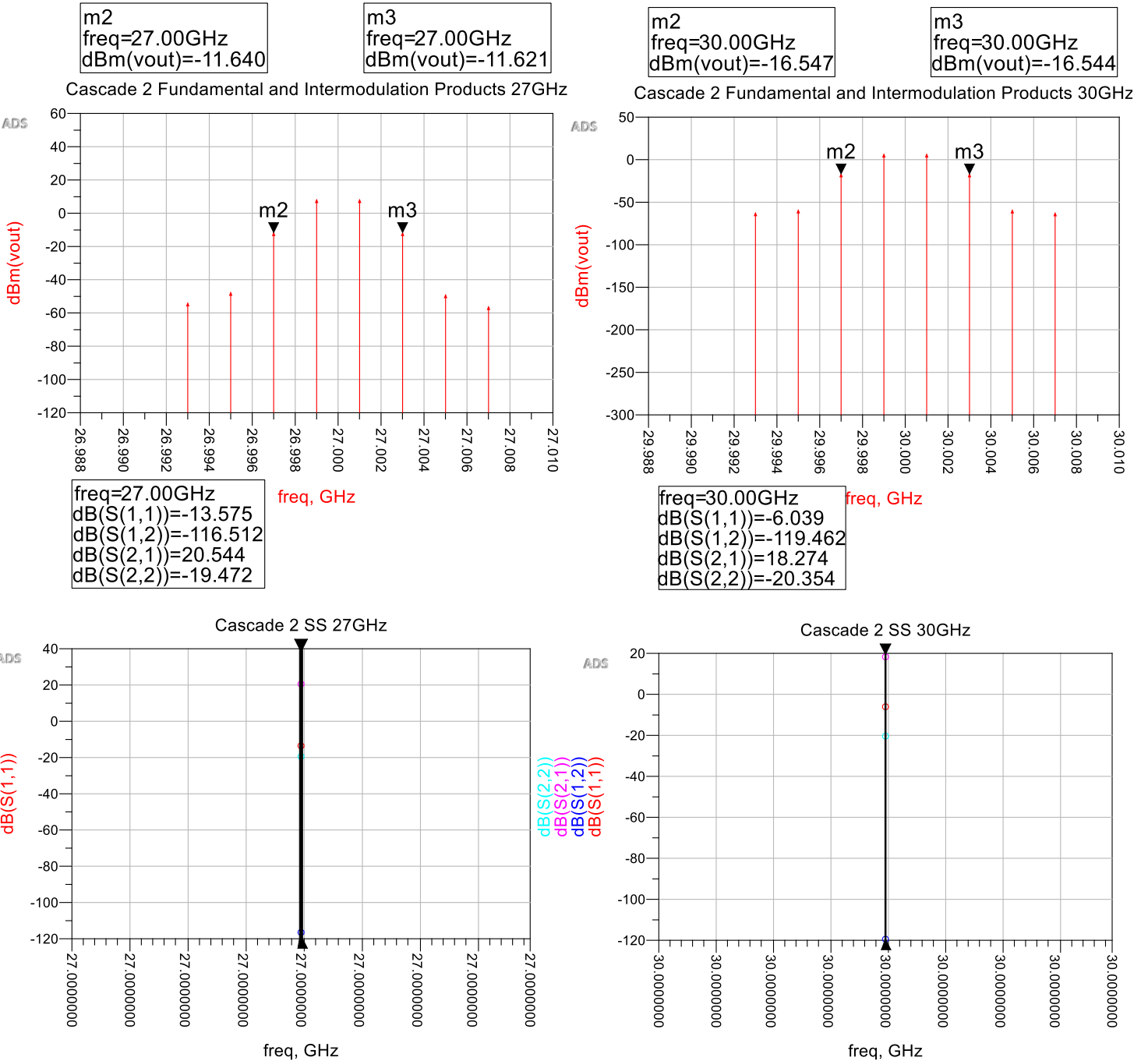


Figure 11: Cascade 3 Schematic

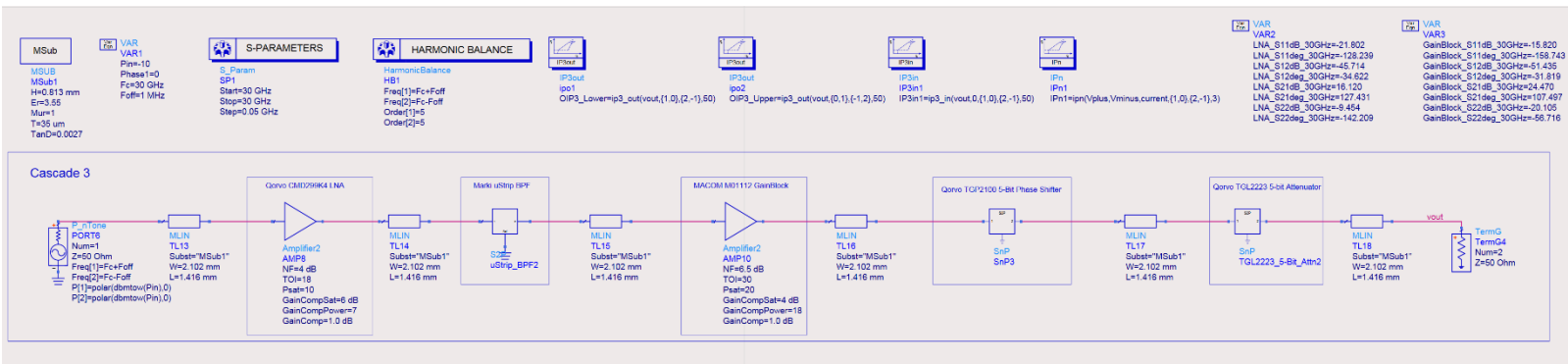


Figure 12: Cascade 3 Third-Order Intercept and Cumulative Gain

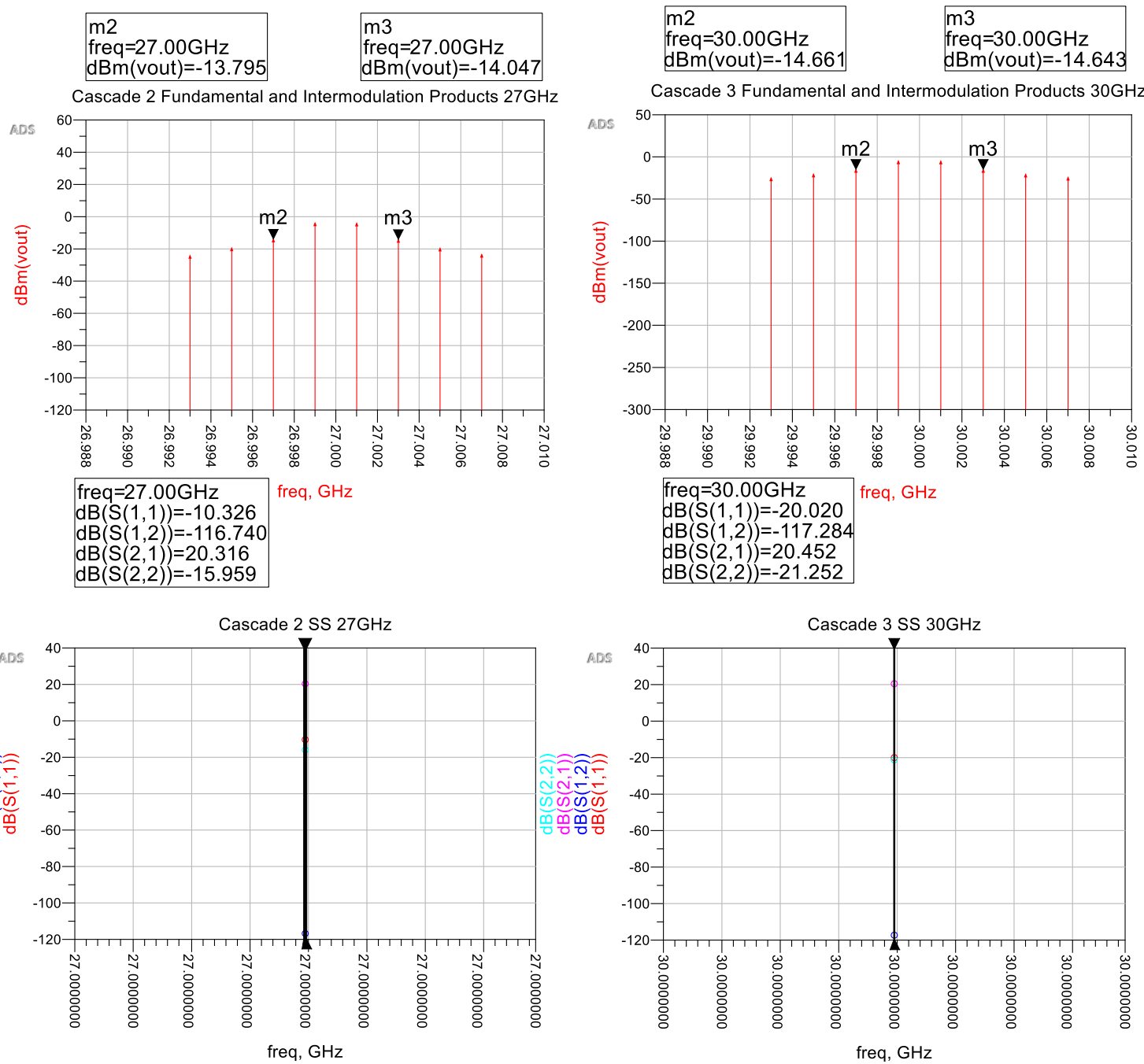


Figure 13: Cascade Passband Gain and Return Loss Comparison

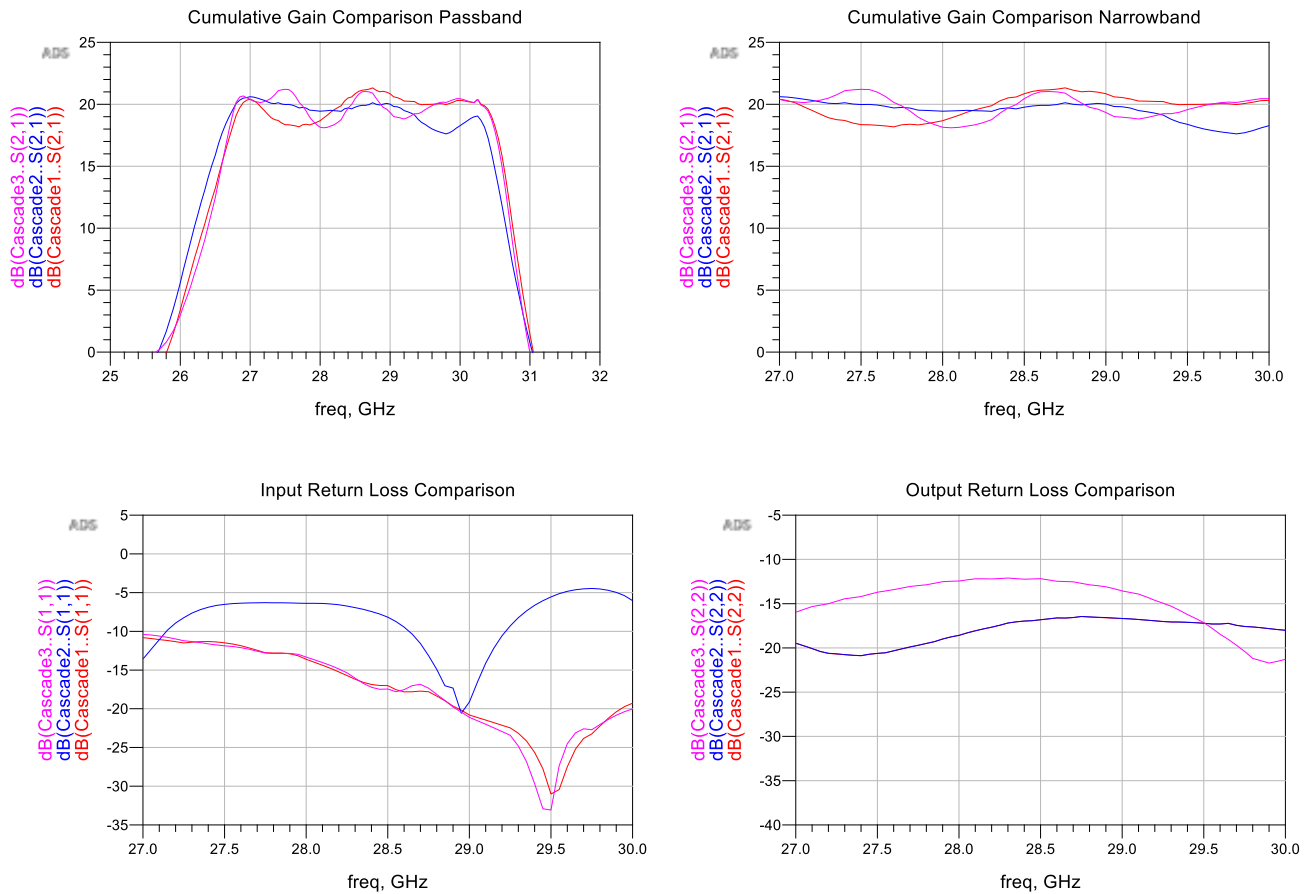


Table 3: Cascade Comparison

Cascade Comparison						
	Cascade 1		Cascade 2		Cascade 3	
	27 GHz	30 GHz	27 GHz	30 GHz	27 GHz	30 GHz
Cumulative NF (dB)	10.63	11.54	10.61	12.26	4.18	4.18
Cumulative Gain (dB)	20.35	20.32	20.54	18.27	20.32	20.45
Σ OTOI (dBm)	18.18	18.05	18.12	18.22	1.27	0.47

Table 4: Calculated vs. Modeled Results

	Cascade 1 Calculated vs. Modeled			
	Calculated		Modeled	
	27 GHz	30 GHz	27 GHz	30 GHz
Cumulative NF (dB)	11.14	10.33	10.63	11.54
Cumulative Gain (dB)	21.23	21.84	20.35	20.32
Σ OTOI (dBm)	22.33	22.83	18.18	18.05
Σ OTOI (W)	0.17	0.19	0.07	0.06
Σ ITOI (dBm)	1.14	1.03	-2.18	-2.27

Cascade 1 was chosen to compare calculated vs. modeled parameters. Calculations were performed using the EXCEL code from question 3.

IV. Conclusions

Comparing the results from the three cascades simulated in ADS, conclusions can be made about arrangement of components by referring to the values in tables 3 and 4. Cascade 1 seems to have the lowest, realistic noise figure of 10.63. Cascade 3 has a noise figure of 4.18 which seems too low for this design. This cascade has the LNA placed at the front connected to the filter and then the gain block with the phase shifter and attenuator following. This is an artifact of the level of detail the model goes to. In order to obtain a more accurate noise figure, noise parameters for all the components would have to be obtained. Doing so would account for the noise produced by the attenuator and phase shifter.

In order to model these cascades more precisely, a 3D electromagnetic simulation would be required. Doing so would account for the microstrip roughness as well as any other electromagnetic phenomena as well as any thermal considerations that may occur when connecting these components together. From figure 14, the passband of the receive chain can be observed. From this, cascade 1 was chosen as the final design with the LNA placed at the front, the Gain Block amplifier at the end, and the filter, variable attenuator, and phase shifter in-between. When comparing the modeled results of cascade 1 from the calculated using the EXCEL code, a slight variation is observed. This is because the calculations are based on equations that make assumptions to mathematically express the gain, noise figure, and third-order intercept of a cascaded system. The simulations performed in ADS produce more accurate results by simulating the transmitted and reflected signals through each component. The impedance mismatches between each component and designed microstrip connections will account for more loss and therefore less gain than calculated. Even with these mismatches accounted for in the calculations, the simulation produces a more precise calculation.

To greater increase the accuracy of this model, large-signal or non-linear models of the components would be required. Using these, as well as simulating the PCB board layers and intrinsic physical properties, more precise models of the cascade could be realized. Based on design requirements an accurate model of a receive chain for a phased array system can be realized. This model can, to some degree, model commercial-off-the shelf components connected by microstrip lines designed for optimal noise figure, third-order intercept, and cumulative gain.

